

Histology

1- Which cell is sensory reception between epidermal cells?

Lymphocyte

Melanocyte

Keratinocyte

✓ Merkel

2- Which type of junctions is an effective barrier to the diffusion of substances across an epithelium?

Gap junctions

Hemidesmosomes

✓ Tight junctions

Desmosomes

3- All of these are located in the chemical synapses, except:

Synaptic vesicles

Postsynaptic receptors

✓ Gap junctions

Mitochondria

4- Which layer of the epidermis has cells which have keratohyaline granules?

Stratum basale

✓ Stratum granulosum

Stratum corneum

Stratum spinosum

5- Which of the following is observed in the histological structure Metaphysis Plate

Osteoclast

Conderoclast

Osteocyte

✓ Chonderocyte

6- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- ☒ Troponin
- ☐ Myosin
- ☐ Tropomyosin
- ☐ Actin

7- In which cell nucleus has cartwheel or clock face arrangement?

- ☐ Mastcell
- ☒ Plasmacell
- ☐ Fibroblast
- ☐ Fibrocyte

8- What is the covering of an individual muscle fiber?

- ☒ Endomysium
- ☐ Sarcolemma
- ☐ Perimysium
- ☐ Epimysium

9- What type of muscles have branching or spindle shape cells?

- ☐ Striated - Smooth
- ☒ Cardiac- Smooth
- ☐ Smooth- Striated
- ☐ Skeletal- Cardiac

10- Structure of framework of spleen and lymphnodes are composed of which fibers?

- ☐ elastin
- ☒ reticular
- ☐ elastic
- ☐ collagen bundles

11- About Unmyelinated PNS axons, which choice is correct?

Oligodendrocyte cell starts to envelop one axon

Schwann cell starts to envelop one axon



Schwann cell starts to envelop multiple axons

Oligodendrocyte cell starts to envelop multiple axons

12- What do you call the simple squamous epithelium that lines the blood vessels?

Pseudostratified

Epithelioid tissue



Endothelium

Mesothelium

13- What is the following is INCORRECT?

cardiac muscle and skeletal muscle have sarcomere



Striation was seen in transverse section of cardiac muscle and skeletal muscle

cardiac muscle and smooth muscle nucleus located in center part

Striation was seen in longitudinal section of cardiac muscle and skeletal muscle

14- Which of the following histological structures has Chondroblast?

Wolkman Canal



Chondrium Matrix

Periosteum

Endosteum

15- Identify the INCORRECT statement

Epithelia are classified by the shape of cell layers

In pseudostratified epithelia all cells are in contact with the basement membrane.

Microvilli increase the surface area of the epithelia cells

The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

16- A nerve is formed through a collection of

Myelin sheet



Axones

Glial cells

Neurons

17- What is the name of macrophage cells in liver?

Kupffer

Osteoclast

Langerhans

Microglial

18- Which cell has a part in formation of blood brain barrier?

Schwann cell

Oligodendrocyte

Ependymal cell



Astrocyte

19- In which cell granules composed of major basic proteins?

Neutrophil



Eosinophil

Basophil

Lymphocyte

20- Which layer of the epidermis contains cells with no nuclei or organelles?

granulosum

spinosum

basale



corneum

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21- An exocrine gland that loses cell bodies during secretion, as is the case for the sebaceous gland, is further classified as a.....gland.

- ☒ Holocrine
- ☐ Apocrine
- ☐ Endocrine
- ☐ Merocrine

22- This type of neuron consist of many dendrites and one true axon , and are more:

- ☒ Multipolar
- ☐ Unipolar or pseudo unipolar
- ☐ Bipolar
- ☐ Apolar

23- In which of the following is Observed only type I collagen ?

- ☒ Perichondrium
- ☐ Elastic Cartilage
- ☐ Hyalin Cartilage
- ☐ Matrix

24- Which cell is the largest white blood cell?

- ☐ Basophil
- ☐ Neutrophil
- ☐ Lymphocyte
- ☒ Monocyte

25- In which of the following is Observed Osteogenic cell &Osteoclast ?

- ☐ Periosteum
- ☐ Pericondrium
- ☐ Nothing
- ☒ Edosteum

26- Which cell secrete antibody in connective tissue?

Fibrocyte



Plasmacell

Mastcell

Adibocyte

27- What is line that bisects the dark band in muscle?

I band



M line

H band

A band

28- In which type of granules in platelets we can find lysosomic enzyme?

Landa

Delta

Beta

Alpha

29- Spinal central canal covered by:

Ependymal cell

Microglia

Oligodendrocyte

Schawann cell

30- Which cell has no nucleus?

Basophil

Normoblast



Erythrocyte

Neutrophil

31- Which of the following is a supporting cell found in the ganglia?

- ✓ Satellite cells
- Neuroglia
- Schwann cells
- Basket cells

32- In which tissue the nucleus are at the rim of the cells?

- Fibroblast
- Macrophage
- Fibrocyte
- ✓ Adipocyte

33- Nissl bodies in the perikaryon of neurons are composed of:

- ✓ Ribosomes and RER
- Golgi and RER
- Ribosomes and Lysosomes
- SER and Lysosomes

34- Which types of neurons are located in dorsal root ganglia?

- Multipolar
- Bipolar
- ✓ Pseudounipolar
- Anaxonic

35- What type of glands are the ceruminous glands in ear?

- Eccrine gland
- ✓ Apocrine gland
- Sebaceous glands
- Endocrine gland

36- Retrograde transportation in the axon is occurred by:

Talin

Kinesin

Actin

✓ Dynein

Embryology

37- Implantation of the blastocyst

cannot occur without prior

differentiation of syncytiotrophoblast

✓ .hatching from the zona pellucida

.fragmentation of the primary yolk sac

development of amnioblast

Embryology

38- A 4-year-old healthy female presents to your office to establish primary care. What phase of meiosis should her oocytes be in?

Meiosis I - prophase

✓

Meiosis II - prophase

Meiosis I - metaphase

Meiosis II – metaphase

39- Maternal serum alpha-fetoprotein increases if the fetus has

oligohydramnios

Bochdalek hernia

. Ladd's bands

✓ anencephaly

40- The intraembryonic coelom

- ✓ it is lined by endoderm
- normally contains only fluid
- becomes the pleural and peritonea
- , but not the pericardial, cavities
- all of the above

41- Pregnancy tests generally are based upon the detection in maternal urine of:

- Estrogen
- Progesterone
- human Chorionic Somatomotropin
- ✓ human Chorionic Gonadotrophin

42- The human placenta

- has a maternal component formed by the decidua capsularis
- shows no changes with age
- retains the cytotrophoblast layer of the placental barrier throughout gestation
- ✓ has chorionic villi as the structural and functional unit

43- As the follicle grows, the oogonium becomes located in a mound of follicle cells called the:

- vesicular antrum
- oocyte
- ✓ cumulus oophorus
- Mural granulosa cells

44- Neural crest

gives rise to most of the cardiogenic mesoderm



is derived from the epiblast

.becomes smooth muscle cells within the wall of the bowel

.induces development of syncytiotrophoblast

45- Which of the following is associated with oligohydramnios

tracheo-esophageal fistula



. renal agenesis

myelomeningocele

anencephal

46- A 3 year-old male patient presents with epicanthal folds, flat nasal bridge, single palmar crease. The patient is 47 XY on karyotype. What could be the cause of this karyotype?

Chromosomal mutation

. Chromosomal inversion



Chromosomal nondisjunction

Chromosomal deletion

47- Somites

are the source of cells in the nucleus pulposus

give rise to cardiac muscle

give rise to the cartilages of the branchial arches



are derived from the paraxial mesoderm

48- Implantation is preceded by an increase in ovarian hCG secretion

- ✓ usually occurs near the end of the first week following fertilization
- requires contact of the zona pellucida with uterine epithelium
- cannot occur outside the uterus and uterine tubes
- leads to the development of the inner cell mass

49- "Quickening" or the period when fetal movements are commonly first felt by the mother occurs:

- near the end of the first trimester
- ✓ around the middle of the second trimester
- at the end of the second trimester
- . at the end of the embryonic period

50- Extraembryonic somatic mesoderm is in close association with:

- syncytiotrophoblast
- ✓ cytotrophoblast & amnion
- yolk sac
- syncytio- & cytotrophoblast

51- The amniotic cavity develops:

- on the tenth day
- ✓ within the inner cell mass near the cytotrophoblast
- in extraembryonic mesoderm
- none of the above

52- The primitive streak

- ✓ is derived from epiblast
- first appears in the roof of the amniotic cavity
- gives rise to ectomesenchyme of the 1st branchial arch
- gives rise to the allantois

53- Major events beginning in the second week of development include:

notochord differentiation

somite formation

angiogenesis

✓ yolk sac development

54- 21 Mesoderm lateral to the notochord:

is derived from hypoblast

✓ differentiates into somite

migrates cranial to prochordal plate

induces endoderm differentiation

55- 22 Which of the following structures does not turn under onto the ventral surface of the embryo during folding of the head?

heart

✓ notochord

pericardial cavity

septum transversum

56- The outer layer of the umbilical cord is continuous with, and similar in structure to the

chorionic membrane

✓ amniotic membrane

syncytiotrophoblast

cytotrophoblast

57- Which layer of the bilaminar embryonic disc forms part of the lining of the amniotic cavity?

epiblast



hypoblast

blastocyst

trophoblast

58- Myelomeningocele (using clinical terminology)

is most common at the tip of the coccyx



leaves the affected individual with paralyzed upper limbs but able to walk

is caused by a failure of the neural folds to meet and fuse

is undetectable by the results of amniocentesis

physiology

59- A motor unit consists of

A motor neuron and whole muscle it innervates.



A motor neuron and the group of skeletal muscle fibers it innervates.

An axonal terminal and whole muscle it innervates.

An axonal terminal and the group of skeletal muscle fibers it innervates.

60- Action potentials spread to the interior of the skeletal muscle fiber by way of

terminal cisternae



transverse tubules

endoplasmic reticulum

longitudinal tubules

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61- Compared to the skeletal muscle fibers, what is the cause of slow cycling of the myosin cross-bridges in the smooth muscle cells?

Low affinity to Ca ions in smooth muscle fibers



Far less ATPase activity of smooth muscle fibers

Higher activity of myosine kinase

Response to different types of stimuli by smooth muscle fibers

62- In which of the following muscle, control is exerted mainly by nerve signals?

Uterus

Ureters

Bile Duct



Ciliary Muscle

63- Which of the following best refers to the importance of chemical synapses?

Synchronous firing of interconnected cells

Rapid firing



Signals amplification

fast conduction velocity

64- During rigor mortis, contraction cycle is stopped at step 3 because of



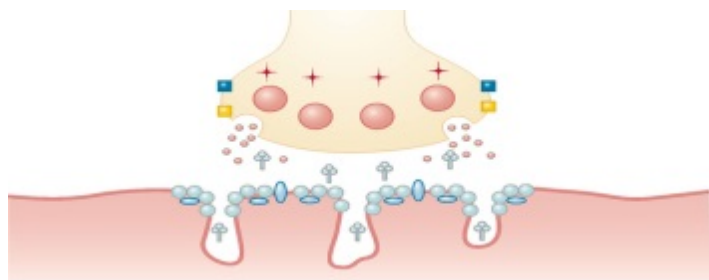
lack of ATP

low affinity of actin filaments to myosin heads

non-functional heads of myosin filaments

ADP and Pi release

65-  In this figure, what does this sign shows?



Acetylcholine molecules

Dens bar

Ca ions

✓ Acetylcholine estrase

66- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

Generation of action potential

Voltage gated sodium channels presense

The amount of resting membrane potential

✓ Velocity of conduction

67- Which of the following represents one of the electrical synapses properties?

Presynaptic vesicles

Chemical transmitter

Synaptic delay

✓ Bidirectional transmission

68- Which of the following is correct about isometric contraction?

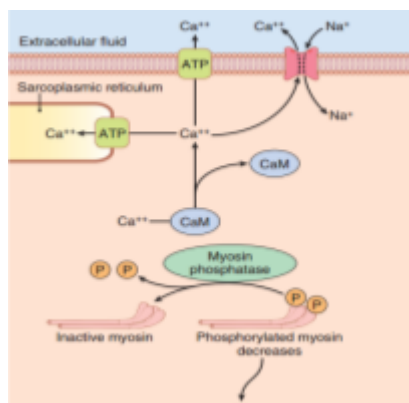
The muscle shortens.

The tension on the muscle remains constant during the contraction.

✓ The load can not be lifted.

It occurs when the force of the muscle contraction is greater than the load.

69-



What does this figure show?

Stress relaxation phenomenon

Smooth muscle contraction

Latch mechanism

Smooth muscle relaxation



70- Which of the following is believed to initiate the contraction process in the skeletal muscle fibers?

Sliding filament mechanism

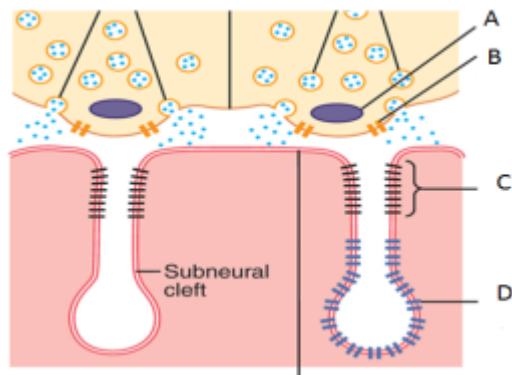
Adenosine triphosphatase activity of the myosin head

Actin-myosin attachment

The strong affinity of the troponin C for calcium ions



71- Which of the following represents "Acetylcholine gated channels"?



A

✓ C

D

B

72- Which of the following is not matched the excitation contraction coupling in the skeletal muscles?

Dihydropyridine receptors

Transverse tubules

Ryanodine receptors

✓ Multiple adherent points

73- In an electromyography report, findings are compatible with severe axonal lesion of a nerve supply to a skeletal muscle. What symptom do you expect to observe in this patient?

Hyperplasia of muscle fibers

✓ Atrophic changes

Remodeling of muscle fibers

Muscle fatigue

74- Which factor inhibits the active sites of actin filaments?

- Myosin heads detachment
- ATP binding to actin filaments
- Inactivation of myosin filaments
- ✓ Tropomyosin-troponin complex

75- In this type of muscle, the source of ca ions needed for contraction is just from sarcoplasmic reticulum?

- Blood vessels
- ✓ Biceps muscle
- Uterus
- Iris muscle of the eye